

WASEDA University
Graduate School of Information, Production and Systems

Diagnosis-Guided Augmentation and Frequency-Gated Fusion for Actor-Conditioned Human-Value Recognition

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Midterm Presentation · Focus: WAIPS-03

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Whose values does a news report describe?

Chile protests: a reported government statement

> *"He justified the decision, saying he wanted to maintain **public order** against vandals and crime."*

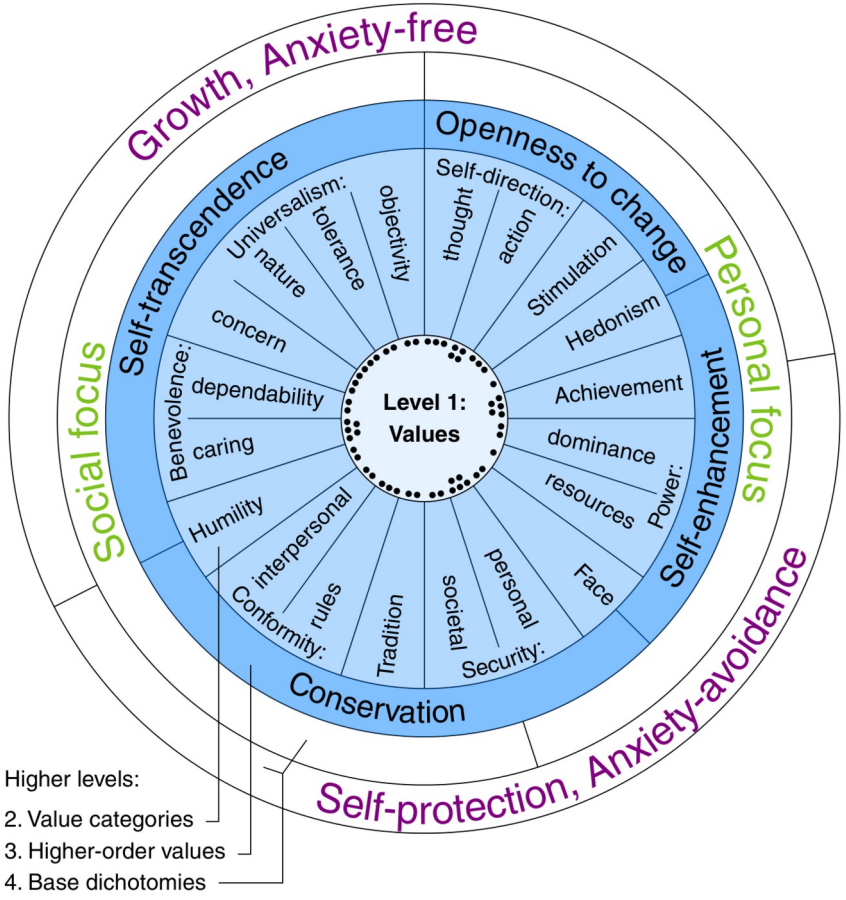
Whose conduct?	Which value?	Which direction?
Government and Security Forces	Have a safe country	**Aligned**

Aligned = supports the value. Contradictory = goes against it.

The benchmark labels reported conduct in its news context.

Source excerpt and reference: Huang, section 5.9, Table 9 (Chile case).

A shared vocabulary for human values



54 specific values

20 broader categories

Values describe what people consider important.

Example: Security, societal

Have a safe country

Have a stable society

Kiesel et al. (ACL 2022),
mainly based on Schwartz's value theory.

Figure: Kiesel et al., ACL 2022, Fig. 1. aclanthology.org/2022.acl-long.306/

NTCIR-19 FEHU: a shared evaluation task

Fact-based Event-centric Human-value Understanding

Values in factual news, at article and event levels

Organizing institutions

University of Tsukuba
Yao Wang, Zhuochen Liu,
Jiankang Chen, Haitao Yu

AIST (Japan)
Xin Liu, Kyoungsook Kim

University of Innsbruck
Adam Jatowt

Shared-task timeline

Date	Milestone
Sep 2025	Task launch
Jan 2026	Training data
Aug 2026	Results returned
8–10 Dec 2026	Conference, Tokyo

Sources: FEHU organizers / important dates; NTCIR-19 conference. Checked 1 Oct 2026.

My task: labels for each supplied actor

Input

News article

Supplied actor list



For each actor

Relevant value + direction

Aligned / contradictory

Several labels may apply

54 values × 2 directions = 108 possible labels

Evidence comes from the article; actor identities are supplied.

Task definition: FEHU official resources; Huang, section 3.1.

Why reliable value labels matter

Potential use	Find relevant news and compare how reports describe different institutions.
Current difficulty	Plausible answers can miss relevant values or add incorrect labels.
Research question	How should supporting predictions guide revision? Which revision steps reduce errors?

Goal: recover relevant labels while controlling incorrect predictions.

Huang, introduction and section 3.3. Uses remain proposed applications.

How the classifier supports the decision

Approach	Role of the trained classifier
EAVIT	Narrow value definitions before the language model decides
WAIPS-03	**Help revise labels after the initial predictions**

Label selection and direction checking use separate decisions.

My design uses classifier scores to support selective revision.

Zhu et al., EAVIT (IJCAI 2025); Huang, section 2. doi.org/10.24963/ijcai.2025/934

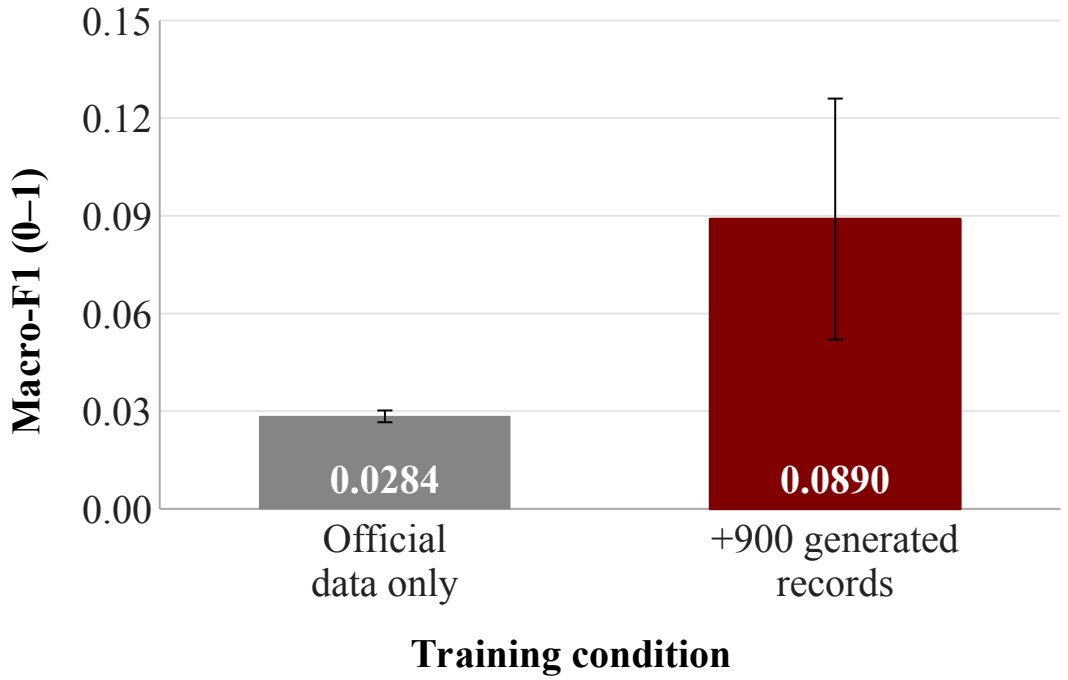
Data augmentation: method and initial evidence

Target 18 missed directed labels

Generate 900 news-like examples linking actors to values and directions.

Train the same DeBERTa classifier under both conditions, with 3 seeds.

Target F1 remains 0.0319 (threshold 0.30).
DRR worsens: 0.1580 to 0.2823.



Three-run means. Error bars: sample SD.

Macro-F1 improves, while the targeted labels remain difficult.

Huang, Table 5, A vs B. Independent training study, distinct from the historical export on slide 16.

WAIPS-02: a simpler comparison system

How it selects labels

Keep a directed label supported by at least 2 of 3 prompts.

Use a separate prediction to choose its direction.

What the comparison tests

Does selective fusion improve recognition

beyond this majority-based approach?

WAIPS-02 provides a simpler reference for the WAIPS-03 results.

Huang, section 4.4: three Pro prompt variants, followed by a Flash direction lookup.

WAIPS-03 revises an initial prediction

PROPOSED METHOD OVERALL PROCESS

For one article and its supplied actors

Initial labels → **Combine and calibrate support** → **Remove or add labels** → **Check disputed directions**

Four auxiliary label sets
108 classifier scores per actor

Frequency and count limits constrain label changes.

Output: a revised set of value–direction labels for each actor.

Additional predictions provide support for selective changes.

Huang, sections 4.5–4.7 and Algorithm 1.

Where the predictions and scores come from

PROPOSED METHOD SOURCES AND OUTPUTS

Source	How the output is produced	Output per actor
Initial E6b	**Flash labels 6–9 generated event descriptions**	**Value + direction**
Auxiliary E1	Fuller examples; label-count caps removed	Value + direction
Auxiliary E6	Flash labels finer event descriptions (median 12)	Value + direction
Auxiliary E6c	E6b descriptions; Pro with thinking disabled	Value + direction
Auxiliary Liu	Fixed targeted few-shot prompt	Value + direction
Classifier	Trained DeBERTa encodes article + actor	108 scores (0–1)

Label sets supply votes and candidates. Classifier scores supply graded support.

Archived E1/E6/E6b/E6c reports; E7 input manifest. Flash/Pro are DeepSeek configurations.

A real candidate recovered by fusion

Devon bus crash: Emergency Services and Authorities

Candidate: Have a safe country — aligned

Prediction source	This candidate
Initial E6b	Absent
Auxiliary E1*	Absent
Auxiliary E6	Aligned
Auxiliary E6c	Aligned
Auxiliary Liu	Aligned

3 exact votes + classifier 0.4399

Raw support ≈ 3.4399

Correctness estimate: 0.4660

Above threshold 0.2697: add aligned.

Final output matches the reference.

Training support: 565 (>130). Value absent; at most two additions per actor.

Supporting predictions recover a value missed by the initial system.

Archived dev case 0-4407-1-1, actor -0. *E1 absence inferred from logged votes; details in notes.

Support scores combine votes and classifier estimates

For one actor and a candidate value–direction label (v, d)

Removal score $s_{rm} = e + 0.25\ o + \max(p_0, p_1)$

Addition score $s_{add} = e + p_d$

e: auxiliary votes for the exact value and direction

o: votes for the same value with only the opposite direction

p_0, p_1: classifier estimates for the two directions. Four auxiliary predictions vote.

Removal accepts partial value support. Addition uses exact direction support.

Huang, §4.6, support-score equations. Actor and value indices omitted for readability.

Calibrated decisions and direction checks

$$q_{rm} = C_{rm}(s_{rm}) \qquad q_{add} = C_{add}(s_{add})$$

C: calibration from development labels. q: estimated correctness.

Remove when $q_{rm} < \theta$

Preserve at least 1 initial label.

Add when $q_{add} \geq \theta$

Absent value, at most 2 labels.

Eligibility: $n_\ell > 130$ training positives. $\theta = F1_{fit}(\text{initial}) / 2$.

Direction QA: one current direction + opposite auxiliary support.

Frequency and count limits constrain revision. The threshold is a heuristic.

Huang, §§4.6–4.7 and Algorithm 1. n is label frequency. Full implementation: backup 25.

Evaluation: recognition and direction errors

Articles: 1,228 training 177 development 175 official test

$$\text{Micro-F1} = 2TP / (2TP + FP + FN)$$

TP: correct, FP: extra,
FN: missed labels

$$\text{Macro-F1} = (\sum_{\ell \in L} F1_{\ell}) / |L|$$

L: labels with gold examples
Each label has equal weight.

$$DRR = R / |E|$$

E: pairs with one gold direction
R: number reversed within E

Higher F1 means better recognition. Lower DRR means fewer reversals.

Huang, §5.2, Micro-F1, gold-supported Macro-F1 and direction reversal rate (DRR).

Recognition scores across four approaches

EVALUATION · TASK 1b, WITH A REPORTED PURE-PROMPT REFERENCE

Method	Micro-F1 ↑	Macro-F1 ↑	DRR ↓
Augmented classifier (historical)	0.3306	0.1303	0.2880
Pure prompt*	0.3787	0.2754	—
WAIPS-02	0.5560	0.3816	0.0338
WAIPS-03: proposed fusion	**0.5817**	**0.4308**	**0.0419**

* Pure prompt: author-reported external score. — = DRR not provided.

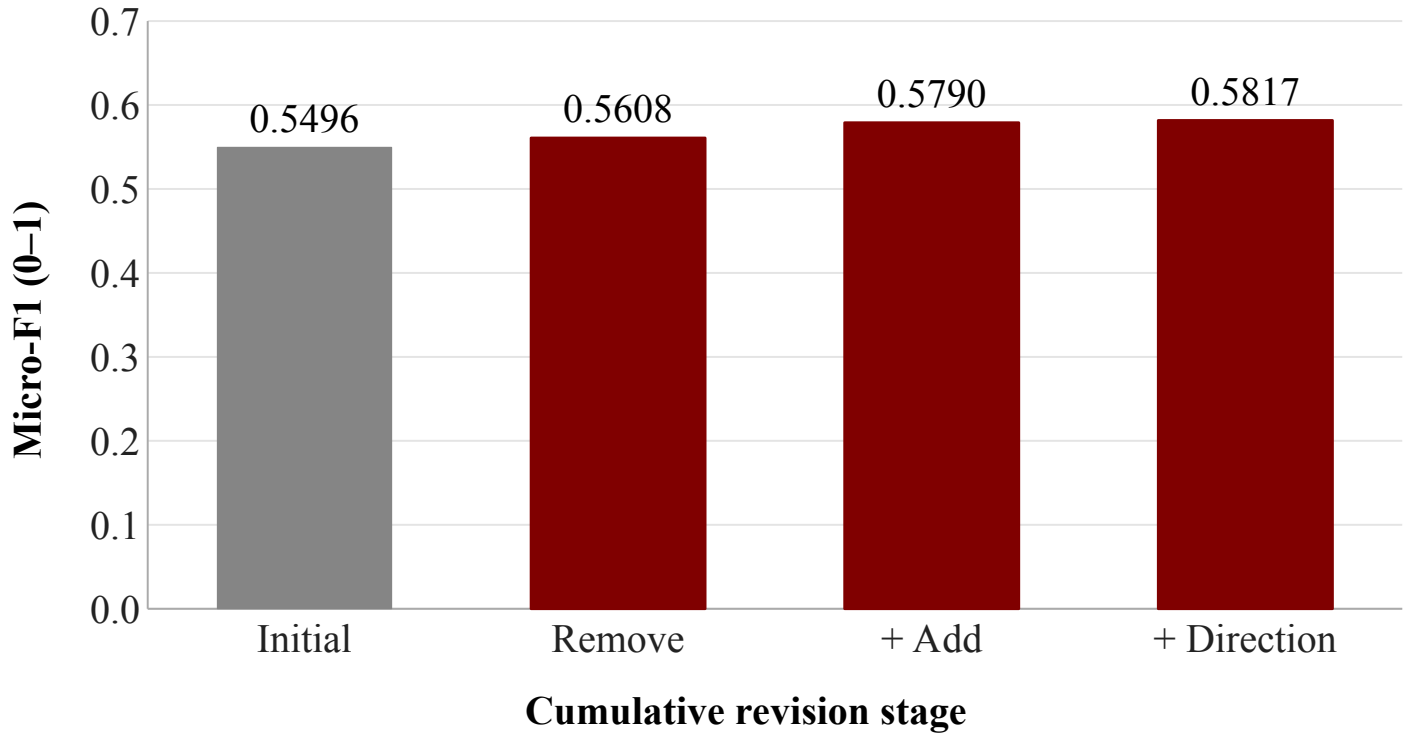
Other rows: 177 dev articles. Encoder: 456 actors, WAIPS: 459. Scores: 0–1.

Fusion improves F1 over WAIPS-02, with a higher reversal rate.

Huang, Table 4 and archived metrics. Pure-prompt F1 provided by Peiyang Huang on 1 Oct 2026.

Which revisions bring the improvement?

EVALUATION · FIXED INITIAL OUTPUTS ON 177 DEVELOPMENT ARTICLES



−170

unmatched labels

−50

missed labels

Net changes from initial

Removal and addition provide most of the gain. Direction checking adds little.

Huang, Table 7, G0/G1/G3/G4. Net label counts from the same saved predictions and development gold.

Official test: better F1, remaining direction errors

EVALUATION · SEPARATE TEST SET, 175 ARTICLES

Submitted system	Micro-F1 ↑	Macro-F1 ↑
WAIPS-02: majority and direction choice	0.5496	0.3580
WAIPS-03: proposed fusion	**0.5808**	**0.3952**

Direction reversal rate rises: 0.0395 to 0.0477.

More predictions oppose the reference direction.

Overall recognition improves, while direction reliability remains a priority.

Official FEHU overview, Table 6; Huang, Table 8. Scores 0–1; lower reversal rate is better.

A direction check can introduce an error

Actor: Government and Security Forces · Value: Have a safe country

> *"...he wanted to maintain **public order** against vandals and crime."*

Before checking	After checking	Reference
Aligned	Contradictory	Aligned

The saved answer contains the direction, without an explanation.

Direction revision can overturn a correct initial judgment.

Huang, section 5.9, Table 9: Chile, 0-4716-3-1, actor -1, value 26. Excerpt shortened.

Next: evidence for more reliable direction changes

FUTURE WORK · PLANNED VALIDATION

A proposed change should cite the actor's conduct

Compare useful corrections and introduced errors
with their supporting passages.

Test on separate evaluation articles

Measure harmful revisions and missed corrections,
alongside overall label recognition.

The aim is fewer harmful changes while keeping useful corrections.

Planned work, extending Huang, sections 6.1–6.2. Effectiveness is not yet established.

Beyond FEHU: value analysis across news sources

FUTURE WORK · PLANNED TRANSFER AND APPLICATION

Research step	Question to test
New sources and events	Does revision still help on news outside the FEHU collection?
Researcher review	Do labels and supporting passages help compare actors' conduct?
Human evaluation	Are labels and passages correct? Does review take less time?

Test transfer to new material and usefulness for human analysis.

Proposed research extension based on Huang, §6.2 evidence evaluation; no transfer result yet.

Conclusion: selective fusion improves recognition

Research process

Training augmentation and prompt aggregation inform WAIPS-03.

Fusion combines classifier scores with auxiliary predictions.

Main result

Higher development and official-test F1 than WAIPS-02

Fewer unmatched and missed labels than the initial prediction

Next: stronger text evidence and evaluation beyond FEHU.

Huang, Tables 4 and 7–9. Main talk ends here.